

2 — Cells, Viruses and Reproduction of Living Things

Edexcel B A-level Biology (9BI0), Topic 2. This topic covers the fine structure of prokaryotic and eukaryotic cells, how microscopes reveal it, the structure and replication of viruses, the cell cycle, mitosis and meiosis (and what happens when chromosomes go wrong), and sexual reproduction in mammals and flowering plants. It contains three core practicals: CP2 (microscopy and micrometers), CP3 (root tip squash) and CP4 (pollen tube growth).

2.1 Eukaryotic and Prokaryotic Cell Structure and Function

2.1 (i) Cell Theory

Cell theory: The idea that every living organism is made of one or more cells, that the cell is the basic unit of structure, function and organisation in living things, and that new cells only arise by the division of existing cells.

Cell theory is described as a **unifying concept** because it applies to every organism, from a single bacterium to a blue whale. It developed over roughly two centuries as microscopes improved: Robert Hooke described and named "cells" in cork in 1665; Schleiden (1838, plants) and Schwann (1839, animals) proposed that all organisms are built from cells; and Virchow (1855) popularised the principle that cells come only from pre-existing cells. It is a good example of how scientific ideas change as new technology produces new evidence.

Viruses sit outside cell theory: they are not made of cells and cannot carry out metabolism or reproduce on their own (see 2.2).

2.1 (ii) Levels of Organisation

In multicellular organisms, cells become **specialised** (differentiated) for particular roles, and specialised cells are grouped into higher levels of organisation:

Level	Meaning	Animal example	Plant example
Cell	Basic unit of life, specialised for a function	Ciliated epithelial cell	Palisade mesophyll cell
Tissue	A group of similar cells (plus any intercellular material) working together to carry out a particular function	Ciliated epithelium lining the trachea	Xylem; palisade mesophyll
Organ	A structure made of several tissues working together to perform one or more functions	Lung; heart	Leaf; root
Organ system	A group of organs working together to carry out a major function	Gas exchange system; circulatory system	Shoot system

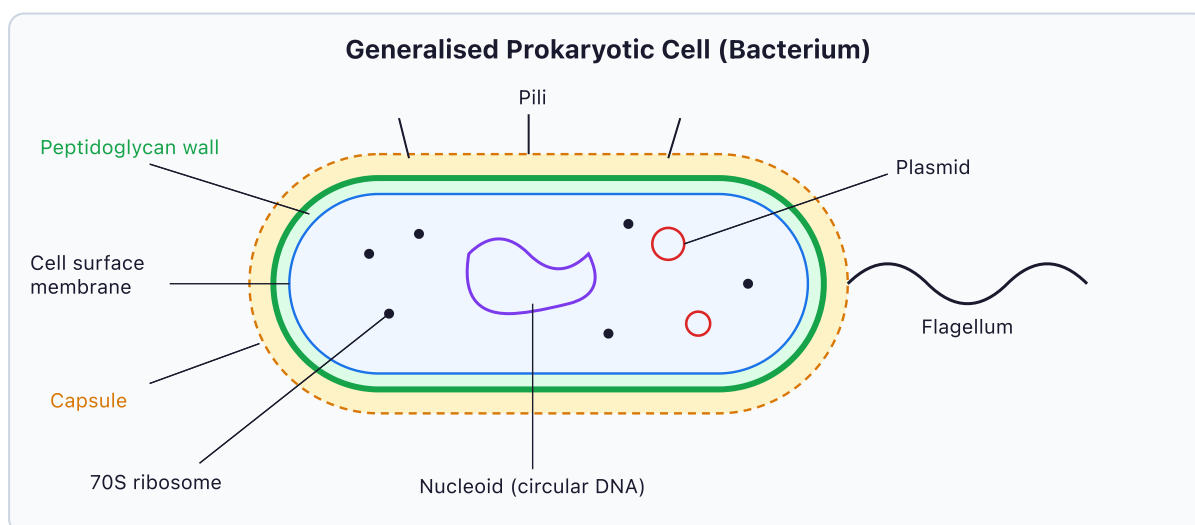
Organisation into tissues and organs allows division of labour: each cell type can be highly efficient at one job, but the cells depend on each other and on transport systems to supply them.

2.1 (iii) Prokaryotic Cell Ultrastructure

Prokaryotes (bacteria and archaea) are usually 0.5–5 μm long, roughly ten times smaller than a typical eukaryotic cell. They have no nucleus and no membrane-bound organelles, but they carry out all the processes of life.

Structure	Description	Function
Nucleoid	Region of cytoplasm containing a single, circular DNA molecule; not surrounded by a membrane; DNA not wound around histones in the way eukaryotic DNA is	Carries the genes needed for the cell to survive and reproduce
Plasmids	Small, circular DNA molecules separate from the main chromosome; replicate independently	Carry extra genes (e.g. antibiotic resistance); can be passed between

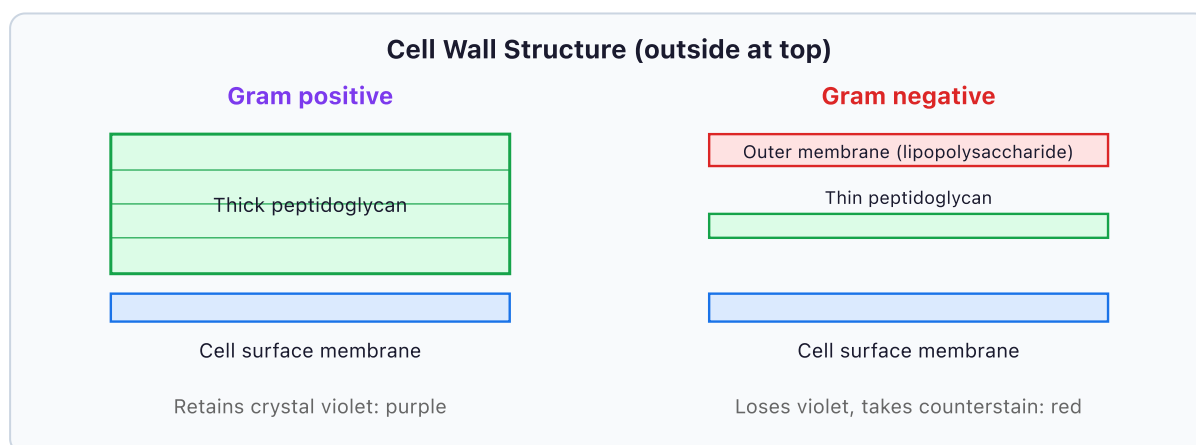
		bacteria; used as vectors in genetic engineering
70S ribosomes	Smaller than eukaryotic ribosomes; made of rRNA and protein (50S + 30S subunits)	Protein synthesis (translation)
Cell wall	Rigid layer outside the cell membrane made of peptidoglycan (murein) — polysaccharide chains cross-linked by short peptides	Supports the cell and stops it bursting when water enters by osmosis
Cell surface membrane	Phospholipid bilayer with proteins	Controls what enters and leaves; site of respiration enzymes (no mitochondria)
Capsule (some species)	Slimy polysaccharide layer outside the wall	Protects against drying out and phagocytosis; helps bacteria stick to surfaces
Flagellum (some species)	Long protein filament rotated at its base	Movement
Pili (some species)	Short protein projections from the surface	Attachment to host cells or surfaces; sex pili are used to transfer plasmids



2.1 (iv) Gram Positive and Gram Negative Cell Walls

The **Gram stain** (devised by Hans Christian Gram in 1884) sorts bacteria into two groups according to the structure of their cell walls. The smear is stained with crystal violet, treated with iodine (which forms a large complex with the dye), washed with alcohol, then counterstained with a red dye such as safranin.

Feature	Gram positive	Gram negative
Peptidoglycan layer	Thick (many layers)	Thin (one or a few layers)
Outer membrane	Absent	Present outside the peptidoglycan; contains lipopolysaccharide
Effect of alcohol wash	Crystal violet–iodine complex is trapped in the thick wall	Alcohol disrupts the outer membrane and the thin wall cannot hold the complex, so the violet washes out
Colour seen	Purple / violet	Red / pink (takes up the counterstain)
Examples	<i>Staphylococcus aureus</i> , <i>Streptococcus</i>	<i>Escherichia coli</i> , <i>Salmonella</i>



Why they respond differently to antibiotics: Penicillin (and other β -lactam antibiotics) stops the enzymes that build the peptide cross-links in peptidoglycan. As a growing bacterium makes new wall, the wall becomes weak; water enters by osmosis and the cell bursts (osmotic lysis). In Gram positive bacteria the peptidoglycan is exposed on the surface, so penicillin reaches its target easily. In Gram negative bacteria the **outer membrane acts as a barrier**, keeping many antibiotics (including penicillin G) and the enzyme lysozyme away from the peptidoglycan, so Gram negative bacteria are generally less

sensitive. Some antibiotics, such as tetracyclines, act on the 70S ribosome instead and work on both groups. Knowing a pathogen's Gram type therefore helps doctors choose an effective drug quickly.

Exam Tip: A common mistake is to say that Gram negative bacteria have "no peptidoglycan" or that "the thick wall stops penicillin". It is the other way round: Gram positive bacteria have a thick, exposed peptidoglycan wall that penicillin can attack; Gram negative bacteria have a thin peptidoglycan layer *protected* by an outer membrane. Penicillin does not harm human cells because they have no peptidoglycan.

2.1 (v) Eukaryotic Cell Ultrastructure

Eukaryotic cells (animals, plants, fungi and protocists) are typically 10–100 µm across. Their DNA is in a nucleus and they contain membrane-bound organelles, which divide the cell into compartments so that different reactions can happen in different conditions at the same time.

Organelle	Structure	Function
Nucleus	Surrounded by a double membrane (nuclear envelope) with nuclear pores; contains chromatin (DNA wound round histone proteins)	Stores genetic information; controls the cell by regulating which genes are transcribed; site of DNA replication and transcription
Nucleolus	Dense, darkly staining region inside the nucleus; not membrane-bound	Makes ribosomal RNA and assembles ribosome subunits
80S ribosomes	Two subunits (60S + 40S) of rRNA and protein; free in the cytoplasm or attached to RER	Translation of mRNA into polypeptides
Rough endoplasmic reticulum (RER)	System of flattened membrane sacs (cisternae) studded with ribosomes; continuous with the nuclear envelope	Proteins made on its ribosomes enter the lumen, where they are folded and transported, often to the Golgi apparatus

Smooth endoplasmic reticulum (SER)	Network of membrane tubules with no ribosomes	Makes and processes lipids, including phospholipids and steroids such as cholesterol
Golgi apparatus	Stack of flattened, curved membrane sacs with vesicles budding off	Modifies proteins (e.g. adds carbohydrate to make glycoproteins), packages them into vesicles for secretion or for use in the cell; makes lysosomes
Mitochondria	Double membrane; inner membrane folded into cristae; matrix contains enzymes, circular DNA and 70S ribosomes	Site of the link reaction, Krebs cycle and oxidative phosphorylation in aerobic respiration, making ATP
Centrioles	Pair of hollow cylinders of microtubules at right angles to each other; found in animal cells but not in flowering plant cells	Organise the spindle fibres during nuclear division; form the bases of cilia and flagella
Lysosomes	Small vesicles bounded by a single membrane, containing hydrolytic enzymes	Digest material taken in by phagocytosis, break down worn-out organelles, and can release enzymes outside the cell
Cell wall (plants)	Layer of cellulose microfibrils outside the membrane; freely permeable	Supports the cell, resists turgor pressure so it does not burst, gives the plant strength
Chloroplasts (plants, algae)	Double membrane (envelope); stacks of thylakoids called grana in a fluid stroma; circular DNA, 70S ribosomes, starch grains	Site of photosynthesis: light-dependent reactions on the thylakoid membranes, light-independent reactions in the stroma
Vacuole (plants)	Large permanent sac filled with cell sap	Stores water, ions, sugars and pigments; keeps the cell turgid, which supports non-woody plants

Tonoplast	The partially permeable membrane surrounding the vacuole	Controls movement of substances into and out of the vacuole
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How organelles work together — the secretory pathway: genes for an extracellular enzyme are transcribed in the nucleus; mRNA leaves through a nuclear pore; ribosomes on the RER translate it and the polypeptide enters the RER lumen; it is carried in a vesicle to the Golgi apparatus and modified; a secretory vesicle then moves to the cell surface membrane and fuses with it, releasing the protein by exocytosis. Mitochondria supply the ATP for this movement and synthesis.

Feature	Prokaryotic cell	Eukaryotic cell
Typical size	0.5–5 μm	10–100 μm
DNA	Circular, in a nucleoid, plus plasmids	Linear chromosomes, with histones, in a nucleus
Ribosomes	70S	80S in cytoplasm (70S inside mitochondria and chloroplasts)
Membrane-bound organelles	None	Present
Cell wall	Peptidoglycan	Cellulose in plants, chitin in fungi, none in animals

Exam Tip: When asked to *identify* organelles in an electron micrograph, use features you can see: cristae (mitochondrion), grana (chloroplast), ribosome-covered cisternae (RER), a curved stack of sacs with vesicles (Golgi). Do not say that plant cells lack mitochondria, or that ribosomes are membrane-bound — both are frequent errors.

2.1 (vi) Magnification and Resolution: Light and Electron Microscopy

Magnification: How many times larger the image is than the real object. **Resolution:** The smallest distance between two points at which they can still be seen as two separate points. Resolution

sets how much detail can be seen; magnifying beyond the resolution limit just gives a bigger, blurred image.

$$\text{magnification} = \text{image size} \div \text{actual size} \quad \text{so} \quad \text{actual size} = \text{image size} \div \text{magnification}$$

Resolution is limited by the wavelength of the radiation used: the shorter the wavelength, the smaller the details that can be separated. Visible light has wavelengths of roughly 400–700 nm, which limits a light microscope to a resolution of about 200 nm (0.2 μm). A beam of electrons has a far shorter wavelength, so electron microscopes can resolve structures well below 1 nm in principle; in biological specimens the practical limit is a few nanometres.

Feature	Light microscope	Transmission electron microscope (TEM)	Scanning electron microscope (SEM)
How the image forms	Light passes through the specimen and is focused by glass lenses	Electrons pass through a very thin section and are focused by electromagnets	Electrons are scanned across the surface and those bounced or emitted from the surface are detected
Useful magnification	Up to about $\times 1500$	Up to about $\times 500\,000$ or more	Up to about $\times 100\,000$ or more
Resolution	About 200 nm	About 0.5 nm or better	A few nm (lower than TEM)
Specimen	Living or dead; can show colour	Dead; fixed, dehydrated, thin-sectioned, in a vacuum	Dead; coated with a thin metal layer, in a vacuum
Image	Colour, 2D	Black and white, 2D, shows internal ultrastructure	Black and white, 3D appearance of surfaces

Worked example (changing the subject; units): A mitochondrion is 36 mm long in a micrograph and is actually 12 μm long. Convert to the same unit: 36 mm = 36 000 μm . Magnification = $36\,000 \div 12 = \mathbf{\times 3000}$. Remember 1 mm = 1000 μm and 1 μm = 1000 nm.

Order of magnitude: a cell 20 μm across is 20 000 nm, which is 200 times a 100 nm virus — a difference of about two orders of magnitude (10^2). Sizes are often written in standard form, e.g. 2×10^{-5}

m for the cell and 1×10^{-7} m for the virus.

2.1 (vii) Why Specimens Are Stained

Most cell components are colourless and nearly transparent, so they give very little contrast. Stains bind selectively to particular chemicals or structures, making them visible and allowing different parts to be told apart.

- **Light microscopy:** iodine solution stains starch blue-black; methylene blue stains nuclei in animal cells; toluidine blue or acetic orcein stain DNA, making chromosomes visible in root tip squashes; the Gram stain distinguishes bacterial wall types. Differential stains colour different tissues differently.
- **Electron microscopy:** specimens are treated with compounds of heavy metals (e.g. lead or uranium salts, osmium tetroxide) that scatter electrons, so stained structures such as membranes appear dark.

Staining usually requires fixing, which kills the cells, and the preparation can introduce **artefacts** (structures that are not present in the living cell).

Core Practical 2 (CP2): Use of the light microscope, including calibrating an eyepiece graticule with a stage micrometer, and drawing a few cells from a specialised tissue. For the full method, see [Core Practicals — CP2](#).

Calibration example: A stage micrometer 1 mm long is divided into 100 divisions, so each division is 10 μm . At $\times 400$, 25 eyepiece divisions line up exactly with 10 stage divisions (100 μm). One eyepiece division = $100 \div 25 = 4 \mu\text{m}$ at this magnification. A cell that spans 12 eyepiece divisions is $12 \times 4 = 48 \mu\text{m}$ long. The graticule must be recalibrated for every objective lens.

Exam Tip: Biological drawings should use a sharp pencil, clear continuous lines with no shading, correct proportions, ruled label lines that do not cross, and a title with the magnification. Only draw what you can see. If a question says *calculate*, show every step, including the unit conversion; if it says *explain*, you need to give reasons (e.g. why electron microscopes resolve finer detail: shorter wavelength).

2.2 Viruses

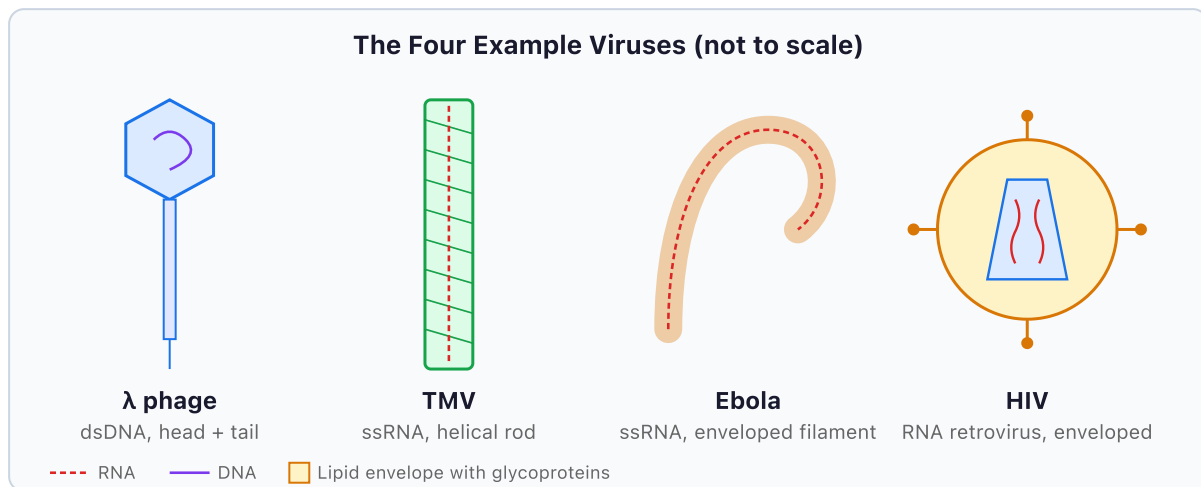
2.2 (i) Classification of Viruses

Virus: A non-cellular infectious particle, usually 20–300 nm across, made of a nucleic acid genome (DNA or RNA) inside a protein coat (**capsid**), sometimes surrounded by a lipid **envelope** taken from the host cell membrane. Viruses can only replicate inside a living host cell.

Viruses are classified mainly by **the type of nucleic acid** they carry (DNA or RNA; single- or double-stranded; whether they use reverse transcription) and by their **structure** (shape of the capsid, and whether there is an envelope). The four examples you need:

Virus	Nucleic acid	Structure	Host
λ (lambda) phage	Double-stranded DNA	Bacteriophage: polyhedral (icosahedral) head containing the DNA, attached to a long tail; no envelope	The bacterium <i>E. coli</i>
Tobacco mosaic virus (TMV)	Single-stranded RNA	Rigid rod: protein subunits arranged in a helix around the RNA; no envelope	Tobacco and related plants, causing mottled (mosaic) leaves
Ebola virus	Single-stranded RNA	Long, thread-like (filamentous) particle, often curved or branched; enveloped, with glycoprotein spikes	Humans and other primates (fruit bats are thought to be the natural reservoir)
Human immunodeficiency virus (HIV)	Single-stranded RNA (two copies); a retrovirus	Roughly spherical; cone-shaped capsid containing RNA and the enzymes reverse transcriptase and integrase; enveloped, with glycoproteins (gp120) that bind to the CD4 receptor	Humans; infects helper T cells

A **retrovirus** uses reverse transcriptase to make a DNA copy of its RNA genome inside the host cell. This DNA is then inserted into the host chromosome by integrase, where it is called a **provirus**.



2.2 (ii) The Lytic Cycle and Latency

In the **lytic cycle** a virus takes over the host cell, makes many copies of itself and then bursts (lyses) the cell:

1. **Attachment:** proteins on the virus bind to specific receptor molecules on the host cell surface. This is why each virus infects only certain cell types (e.g. HIV binds CD4 on helper T cells).
2. **Entry:** a phage injects its DNA through the bacterial wall, leaving the capsid outside; enveloped animal viruses enter when the envelope fuses with the host membrane or when the cell takes them in by endocytosis.
3. **Replication and synthesis:** the host cell's ribosomes, enzymes, nucleotides, amino acids and ATP are used to copy the viral genome and make viral proteins. The host's own metabolism is disrupted.
4. **Assembly:** new capsids form and are filled with viral nucleic acid.
5. **Release:** the host cell bursts, releasing large numbers of new virus particles (for phages, lysis is caused by a viral enzyme that weakens the wall). Enveloped viruses often leave by budding, taking a piece of host membrane as their envelope.

Latency: A stage in which the viral genome stays inside the host cell without new virus particles being made, so there are no symptoms. In λ phage (the **lysogenic** pathway) the DNA is inserted into the bacterial chromosome as a **prophage** and is copied every time the bacterium divides. In HIV, the provirus sits in the DNA of the host cell. A trigger (such as stress or damage to the host cell) can later switch the virus into the lytic cycle.

Latency explains why some infections can reappear years later (e.g. cold sores from herpes simplex) and why HIV can remain in the body even when drug treatment has made the virus undetectable in blood.

2.2 (iii) Antivirals

Viruses are not cells: they have no cell wall, no 70S ribosomes and no metabolism of their own.

Antibiotics target bacterial structures and processes, so they have no effect on viruses. **Antiviral drugs** must instead stop the virus replicating, usually by blocking a step or enzyme that is unique to the virus, so that host cells are harmed as little as possible. Examples:

- **Reverse transcriptase inhibitors** (e.g. zidovudine, AZT) stop HIV making DNA from its RNA.
- **Protease inhibitors** stop HIV cutting its proteins into their working forms, so new particles are not infectious.
- **Neuraminidase inhibitors** (e.g. oseltamivir) stop new influenza particles being released from the host cell.
- Other antivirals block attachment or entry, or are nucleotide analogues that stop the viral genome being copied (e.g. aciclovir for herpes).

Because viruses use the host's machinery, there are relatively few viral targets, and viruses (especially RNA viruses) mutate rapidly, so resistance to a single drug can develop. HIV is therefore treated with combinations of drugs.

2.2 (iv) Preventing Spread: the 2014 Ebola Outbreak

Once a viral infection is established it can be hard to treat, because there are few effective antivirals and the virus is inside the body's own cells. Disease control therefore concentrates on **stopping transmission**.

The West African Ebola epidemic began in Guinea in late 2013 and spread through 2014 to Sierra Leone and Liberia, the largest Ebola outbreak recorded. By the time it ended in 2016 the World Health Organization had recorded more than 28 000 cases and over 11 000 deaths. Ebola virus disease causes fever, vomiting, diarrhoea and in some patients bleeding; many patients died. The virus spreads through direct contact with the blood or other body fluids of infected people (living or dead), or with contaminated objects. Health workers, family carers and people handling bodies at funerals were at high risk.

Measures used to stop the spread included:

- **Isolating** patients in specialised treatment centres, and quarantining people who had been in contact with them.
- **Contact tracing:** finding and monitoring everyone who had been in contact with a case for the 21-day incubation period.
- **Personal protective equipment** (gowns, gloves, masks, eye protection) and strict decontamination for health workers.
- **Safe burial** teams, because traditional burial practices involving touching the body spread infection.
- **Public education** about how the virus is spread; screening of travellers at airports and borders.

2.2 (v) Ethics of Using Untested Drugs in Epidemics

At the start of the 2014 outbreak there was no licensed drug or vaccine for Ebola. A few experimental treatments were given to a small number of patients on compassionate grounds, and in August 2014 a WHO ethics panel concluded that offering unproven treatments could be ethical if certain conditions were met (such as informed consent, openness about what is known, and collecting data). Later, an experimental vaccine was tested in a large trial in Guinea in 2015.

Arguments for using untested drugs	Arguments against / concerns
Very high death rate, so a small chance of benefit may outweigh unknown risks	Unknown side effects; the drug could harm or even kill patients (non-maleficence)
Patients have a right to choose (autonomy) if they give informed consent	Truly informed consent is hard to obtain from very ill, frightened people, or where language and education are barriers
Gathering data during an outbreak may be the only chance to find out whether a drug works	Without a proper controlled trial it is hard to tell whether a drug works; using a placebo group when many patients die is itself ethically difficult
Could reduce the number of deaths and help end the epidemic	Supplies are very limited: who decides who receives them (justice)? Early doses went mostly to foreign health workers, which raised concerns about fairness
Standard testing takes many years, which is too slow in an emergency	Could draw resources and attention away from proven measures such as isolation and supportive care

Exam Tip: For an *evaluate* or *discuss* question on ethics, give points on both sides, link them to principles (safety, consent, fairness, benefit to society), and finish with a justified conclusion. Also be ready to *explain* why antibiotics cannot treat a viral infection — say which bacterial targets viruses lack.

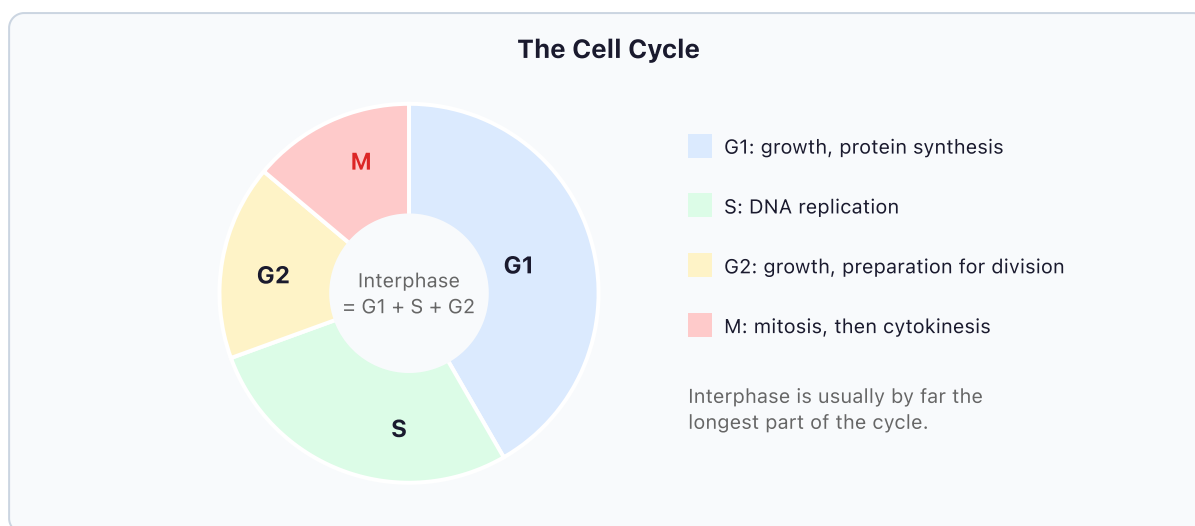
2.3 Eukaryotic Cell Cycle and Division

2.3 (i) The Cell Cycle

Cell cycle: The controlled sequence of events by which a eukaryotic cell grows, copies its DNA and divides to give two genetically identical daughter cells. It has three main stages: **interphase**, **mitosis** (nuclear division) and **cytokinesis** (division of the cytoplasm).

- **G1 (first growth phase):** the cell grows, makes proteins and organelles, and carries out its normal functions.
- **S phase (synthesis):** DNA is replicated semi-conservatively, so each chromosome becomes two identical **sister chromatids** joined at a **centromere**.
- **G2 (second growth phase):** the cell continues to grow and makes the proteins needed for division, such as tubulin for the spindle; organelles such as mitochondria are replicated.
- **Mitosis** then **cytokinesis**.

The cycle is **regulated** at checkpoints (for example at the end of G1 and before mitosis) where the cell checks that it is large enough, that DNA is undamaged and that replication is complete. Cells that stop dividing leave the cycle into a resting state (G0), as many nerve and muscle cells do. When the controls fail, cells may divide uncontrollably and form a tumour.



2.3 (ii) Genetic Material During the Cell Cycle and the Stages of Mitosis

During interphase the DNA is spread out as chromatin, so individual chromosomes cannot be seen. After S phase the amount of DNA per nucleus has doubled (from $2x$ to $4x$), though the chromosome number ($2n$) is unchanged, because each chromosome now has two chromatids. Mitosis shares these chromatids equally so each daughter nucleus returns to $2x$ with the full diploid set.

Stage	What happens
Prophase	Chromosomes condense (supercoil) so they become shorter, thicker and visible, each as two sister chromatids joined at the centromere. The nucleolus disappears and the nuclear envelope breaks down. In animal cells the centrioles move to opposite poles; spindle fibres (microtubules) form between the poles.
Metaphase	Spindle fibres attach to the centromere of each chromosome, and the chromosomes line up individually across the equator of the spindle.
Anaphase	The centromeres split; spindle fibres shorten and pull the sister chromatids apart to opposite poles, centromere first (giving a V shape).
Telophase	The separated chromatids (now chromosomes) reach the poles and uncoil; a nuclear envelope forms around each set and nucleoli reappear; the spindle breaks down.
Cytokinesis	Animal cells: the membrane is pulled inwards to form a cleavage furrow that pinches the cell in two. Plant cells: vesicles from the Golgi apparatus gather at the equator and fuse to form a cell plate, which becomes the new cell wall.

Exam Tip: Distinguish between a *chromosome*, a *chromatid* and *chromatin*. DNA replication happens in **S phase of interphase**, not in prophase. In metaphase of mitosis, chromosomes line up *singly*; in metaphase I of meiosis they line up in *homologous pairs*. Examiners also penalise "the chromosomes split" — say the centromeres divide and the sister chromatids separate.

2.3 (iii) Roles of Mitosis

- **Growth:** a multicellular organism develops from one zygote by repeated mitosis, all cells having the same genetic information. In plants, growth by mitosis happens in meristems, such as those at root and shoot tips.
- **Repair and replacement:** damaged tissue is repaired and worn-out cells (skin, gut lining, blood cells) are replaced with identical cells that can do the same job.
- **Asexual reproduction:** a single parent produces genetically identical offspring (clones) — for example budding in yeast and *Hydra*, and vegetative propagation in plants such as strawberry runners and potato tubers. This is fast and does not need a mate, but produces no genetic variation, so a whole population may be vulnerable to the same disease or environmental change.

Core Practical 3 (CP3): Prepare a temporary squash of a root tip, stain it and observe the stages of mitosis in the meristem with a light microscope. For the full method, see [Core Practicals — CP3](#).

$$\text{mitotic index} = \text{number of cells in mitosis} \div \text{total number of cells counted}$$

Worked example: In a field of view of a root tip, 24 of 200 cells are in a stage of mitosis. Mitotic index = $24 \div 200 = 0.12$, or 12%. The proportion of cells in each stage is proportional to the time that stage lasts, so if the whole cycle takes 20 hours and 5% of cells are in metaphase, metaphase lasts about $0.05 \times 20 = 1$ hour.

2.3 (iv) Meiosis and Haploid Gametes

Meiosis: A type of nuclear division involving two successive divisions (meiosis I and meiosis II), after a single round of DNA replication, that produces four haploid (n) nuclei that are genetically different from each other and from the parent cell.

Halving the chromosome number is essential for sexual reproduction: when two haploid gametes fuse at fertilisation, the diploid number is restored (in humans, $23 + 23 = 46$) instead of doubling every generation.

Stage	Meiosis I (separates homologous chromosomes)	Meiosis II (separates sister chromatids)
Prophase	Chromosomes condense; homologous chromosomes pair up to form bivalents ; crossing over occurs at chiasmata ; nuclear envelope breaks down; spindle forms	Chromosomes (still two chromatids each) condense again; new spindles form
Metaphase	Bivalents line up on the equator; the orientation of each pair is random	Individual chromosomes line up on the equator
Anaphase	Homologous chromosomes are pulled to opposite poles; centromeres do <i>not</i> divide	Centromeres divide; sister chromatids are pulled to opposite poles
Telophase	Two haploid nuclei form; cytokinesis usually follows	Four haploid nuclei form; cytokinesis gives four cells

Feature	Mitosis	Meiosis
Number of divisions	One	Two
Daughter cells	Two, diploid ($2n$)	Four, haploid (n)
Genetically	Identical to each other and parent	Different from each other and parent
Homologous pairing / crossing over	No	Yes, in prophase I
Used for	Growth, repair, asexual reproduction	Producing gametes for sexual reproduction

2.3 (v) Meiosis and Genetic Variation

Meiosis produces new combinations of alleles (**recombination**) in two ways:

- **Independent assortment** (metaphase I and anaphase I): each homologous pair lines up independently of every other pair, so which maternal and paternal chromosomes end up together in a gamete is random. The number of possible combinations is 2^n , where n is the haploid number. For humans, $2^{23} = 8\,388\,608$ (about 8.4 million) combinations. Independent assortment in meiosis II also adds variation where crossing over has made the sister chromatids different.
- **Crossing over** (prophase I): non-sister chromatids of a homologous pair break and rejoin at chiasmata, exchanging equivalent sections. This produces chromatids carrying new combinations of alleles of linked genes (genes on the same chromosome).

Random fertilisation then combines two such gametes, so the number of possible genetic combinations in offspring is enormous. This variation is the raw material for natural selection.

Exam Tip: Crossing over produces new combinations of *alleles*, not new alleles — only mutation creates new alleles. Crossing over happens between *non-sister* chromatids of *homologous* chromosomes. When a question asks you to *calculate* gamete combinations from independent assortment, use 2^n with n as the number of homologous pairs.

2.3 (vi) Chromosome Mutations: Translocations

Chromosome mutation: A change in the structure or number of whole chromosomes, large enough in structural cases to affect many genes. **Translocation:** a section of one chromosome breaks off and becomes attached to another, non-homologous chromosome.

Translocations may be **reciprocal** (two non-homologous chromosomes swap sections) or non-reciprocal. In a **balanced** translocation no genetic material is lost or gained, so the carrier is usually healthy, but at meiosis the chromosomes may be shared out unevenly, producing gametes with missing or extra sections. This can cause miscarriage or genetic disorders in offspring. Translocations can also harm the person carrying them if a break point disrupts a gene or joins two genes together:

- In most cases of **chronic myeloid leukaemia**, a reciprocal translocation between chromosomes 9 and 22 produces the shortened "Philadelphia chromosome", carrying a fused gene (*BCR-ABL*)

whose protein continually stimulates cell division.

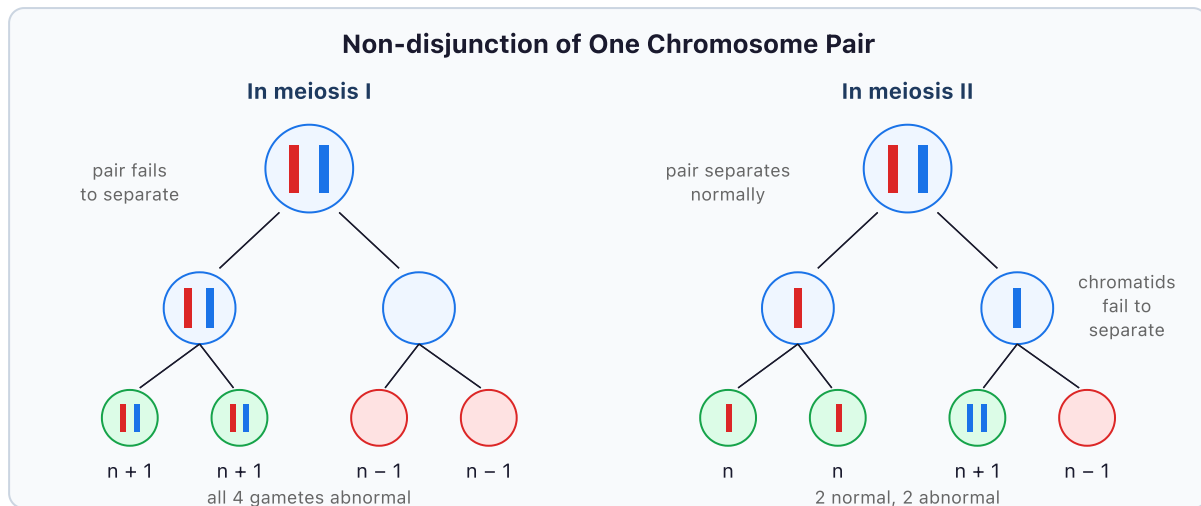
- A small percentage of cases of Down's syndrome are caused by a translocation in which an extra copy of most of chromosome 21 is attached to another chromosome (often chromosome 14).

2.3 (vii) Non-disjunction: Polysomy and Monosomy

Non-disjunction: The failure of homologous chromosomes (in anaphase I) or sister chromatids (in anaphase II) to separate properly during meiosis, so that some gametes have an extra copy of a chromosome ($n + 1$) and some lack one ($n - 1$).

If an $n + 1$ gamete is fertilised by a normal gamete, the zygote has three copies of that chromosome (**trisomy**, a form of **polysomy** — more than two copies of a chromosome). If an $n - 1$ gamete is fertilised, the zygote has only one copy (**monosomy**). Most such zygotes do not survive, but some do:

Condition	Karyotype	Cause	Main features
Down's syndrome (trisomy 21) — polysomy	47 chromosomes, three copies of chromosome 21	Usually non-disjunction of chromosome 21 during meiosis, most often in the mother; the risk rises with maternal age	Learning difficulties, characteristic facial features, higher risk of heart defects
Turner's syndrome — monosomy	45 chromosomes, a single X (45,X)	A gamete lacking a sex chromosome fuses with an X-bearing gamete	Female; short stature; ovaries do not develop normally, so usually infertile



Non-disjunction in meiosis I makes all four gametes abnormal (two $n + 1$, two $n - 1$); in meiosis II, only the two gametes from the affected cell are abnormal (one $n + 1$, one $n - 1$), and the other two are normal. (Each bar in the diagram stands for one chromosome of the pair; the chromatids are not drawn separately.)

Exam Tip: Non-disjunction is a *chromosome number* mutation; a translocation is a *chromosome structure* mutation. When asked to *describe* non-disjunction, name the stage (anaphase I or II), say what fails to separate (homologous chromosomes or sister chromatids), and state the resulting gametes ($n + 1$ and $n - 1$).

2.4 Sexual Reproduction in Mammals

2.4 (i) Spermatogenesis and Oogenesis

Gametogenesis in both sexes follows the same pattern: diploid germ cells multiply by mitosis, grow, and then undergo meiosis to produce haploid cells. The differences lie in timing and in how the cytoplasm is shared.

Spermatogenesis takes place in the walls of the seminiferous tubules of the testes, supported and nourished by Sertoli cells:

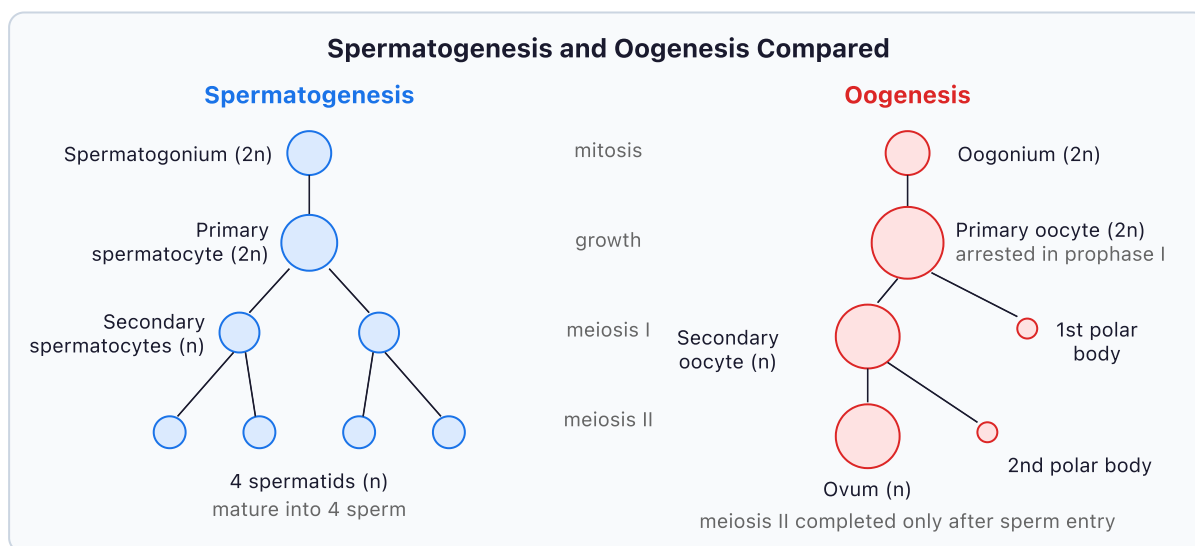
1. Diploid **spermatogonia** near the outer edge of the tubule divide repeatedly by mitosis.
2. Some grow into diploid **primary spermatocytes**.
3. Each primary spermatocyte completes meiosis I to form two haploid **secondary spermatocytes**.

- Meiosis II gives four haploid **spermatids**.
- Spermatids mature into **spermatozoa**, developing an acrosome, a mitochondria-rich midpiece and a flagellum, and they are released into the lumen of the tubule.

Sperm production continues throughout life from puberty onwards, producing millions of sperm every day.

Oogenesis takes place in the ovaries:

- Before birth, diploid **oogonia** in the fetal ovary divide by mitosis and grow into diploid **primary oocytes**, each surrounded by follicle cells (a primary follicle).
- The primary oocytes start meiosis I but stop in prophase I. A baby girl is born with all the primary oocytes she will ever have.
- From puberty, in each menstrual cycle, a follicle develops and its primary oocyte completes meiosis I. The cytoplasm divides unequally, forming a large haploid **secondary oocyte** and a small **first polar body**.
- The secondary oocyte starts meiosis II but stops at metaphase II. It is released from the ovary at ovulation.
- Meiosis II is completed only if a sperm enters, producing the **ovum** and a **second polar body**. Polar bodies eventually degenerate.



Feature	Spermatogenesis	Oogenesis
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Where	Seminiferous tubules of testes	Ovaries (meiosis II completed in the oviduct after fertilisation)
Functional gametes per meiosis	Four sperm	One ovum (plus polar bodies)
Division of cytoplasm	Equal	Unequal, so the ovum keeps most of the cytoplasm and food reserves
Timing	Continuous from puberty	Begins before birth; pauses; one secondary oocyte usually released per cycle from puberty to menopause
Gamete size and motility	Small, motile	Large, not motile

2.4 (ii) Fertilisation

Fertilisation in humans normally takes place in the oviduct. The sequence, from first contact to the fusion of nuclei, is:

1. **Capacitation:** while in the female reproductive tract, chemical changes to the sperm cell surface membrane (such as removal of cholesterol and glycoproteins) make the sperm more active and able to undergo the acrosome reaction.
2. **Passing the follicle cells:** the sperm swims through the layer of follicle cells (corona radiata) around the secondary oocyte.
3. **Acrosome reaction:** when the sperm head binds to receptors on the **zona pellucida** (a glycoprotein layer around the oocyte), the acrosome releases hydrolytic (protease) enzymes by exocytosis. These digest a path through the zona pellucida.
4. **Fusion of membranes:** the sperm cell surface membrane fuses with the oocyte membrane, and the sperm nucleus enters the oocyte cytoplasm.
5. **Cortical reaction:** cortical granules just under the oocyte membrane release their contents by exocytosis into the space beneath the zona pellucida. Enzymes change the zona so that it hardens and its sperm receptors are destroyed, preventing further sperm entering (preventing **polyspermy**).
6. **Completion of meiosis II:** sperm entry triggers the secondary oocyte to finish meiosis II, forming the ovum nucleus and the second polar body.

7. **Fusion of nuclei:** the haploid male and female nuclei move together and fuse, forming a diploid **zygote**.

Exam Tip: Remember the correct order: acrosome reaction *then* membrane fusion *then* cortical reaction. The acrosome reaction happens in the *sperm*; the cortical reaction happens in the *oocyte*. The cell released at ovulation is a secondary oocyte, not a mature ovum. The cortical reaction does not "kill" other sperm — it changes the zona pellucida so that no more can bind or get through.

2.4 (iii) Early Development to the Blastocyst

After fertilisation the zygote moves down the oviduct towards the uterus, dividing by mitosis as it goes. These first divisions are called **cleavage**: the cells do not grow between divisions, so the embryo stays about the same overall size while its cells get smaller.

- The zygote divides to form 2, then 4, then 8 cells, and so on.
- After roughly 3–4 days a solid ball of about 16 or more cells, the **morula**, reaches the uterus.
- By about day 5 fluid collects inside it and a hollow **blastocyst** forms, with three parts:
 - an outer layer of cells, the **trophoblast**, which will help implant the embryo and form part of the placenta;
 - the **inner cell mass**, a cluster of cells at one side that will develop into the embryo (these are embryonic stem cells);
 - a fluid-filled cavity (the **blastocoel**).
- The blastocyst breaks out of the zona pellucida and, about 6–7 days after fertilisation, begins to **implant** in the endometrium (lining) of the uterus.

2.5 Sexual Reproduction in Plants

2.5 (i) Formation of Pollen Grains and the Embryo Sac

Pollen grains (male gametophytes) in the anther:

1. Each anther contains four pollen sacs filled with diploid **pollen mother cells**.
2. Each pollen mother cell divides by meiosis to form a **tetrad** of four haploid cells (microspores).

3. The tetrad cells separate and each develops a tough, resistant outer wall (**exine**, made of sporopollenin) and a thin inner wall (**intine**).
4. The haploid nucleus divides by **mitosis** to give a **tube nucleus** and a **generative nucleus**. The pollen grain is now mature.
5. The anther wall dries and splits (dehiscence), releasing the pollen.

The embryo sac (female gametophyte) in the ovule:

1. Inside each ovule, within the nucellus and surrounded by the integuments (which leave a small gap, the **micropyle**), a diploid **megaspore mother cell** divides by meiosis, producing four haploid cells.
2. Three of these degenerate; the surviving **megaspore** enlarges.
3. Its nucleus divides by mitosis three times, giving **eight haploid nuclei**.
4. These become arranged as: the **egg cell** and two **synergids** at the micropyle end; three **antipodal cells** at the opposite end; and two **polar nuclei** in the centre (in the central cell). This is the mature embryo sac.

Exam Tip: In flowering plants meiosis produces spores, and it is *mitosis* that then produces the gamete nuclei. Do not write that the generative nucleus or egg cell is made by meiosis. The pollen grain is not itself a gamete; the male gametes are the two nuclei formed from the generative nucleus.

2.5 (ii) How the Male Nuclei Reach the Embryo Sac

1. **Pollination** transfers pollen from an anther to a compatible stigma (by insects or wind, for example).
2. The pollen grain absorbs water and sugars from the sticky stigma surface and **germinates**: a **pollen tube** grows out through a pore (aperture) in the exine.
3. The tube grows down through the style. Its tip secretes **digestive (hydrolytic) enzymes** which break down the tissues of the style; the products of digestion are absorbed and used for growth of the tube.
4. The **tube nucleus** stays near the growing tip and is thought to control the growth of the tube, by controlling the production of the enzymes and proteins needed.
5. The tube grows towards the ovule, guided by chemical signals released from the ovule (chemotropism).

6. As it grows, the **generative nucleus divides by mitosis** to form **two male nuclei** (male gametes), which follow the tube nucleus down the tube.
7. The tube enters the ovule through the micropyle; the tube nucleus degenerates and the tip of the tube bursts, releasing the two male nuclei into the embryo sac.

Core Practical 4 (CP4): Investigate how sucrose concentration affects pollen tube growth or germination. For the full method, see [Core Practicals — CP4](#).

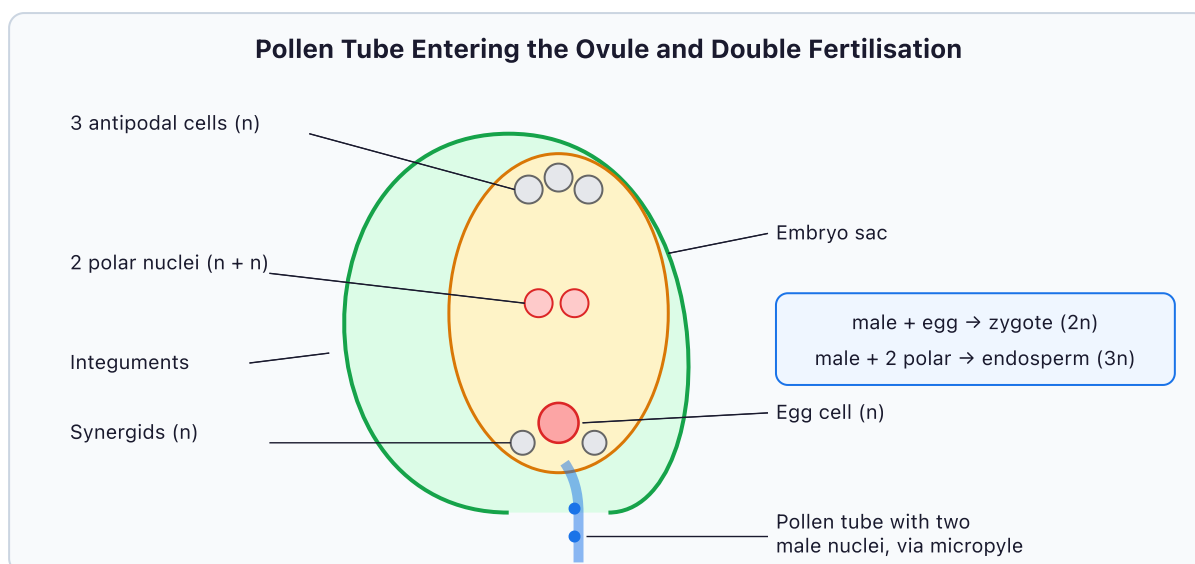
In CP4, the sucrose concentration is the independent variable; pollen tube length (or the percentage of grains that have germinated) is the dependent variable, measured using a calibrated eyepiece graticule after a fixed time. Plotting the two variables on a graph shows the trend. Sucrose supplies respiratory substrate and affects the water potential of the medium, so if the concentration is too high the pollen loses water by osmosis and tube growth falls.

2.5 (iii) Double Fertilisation

Double fertilisation: The process, unique to flowering plants, in which two separate fusions take place in the embryo sac, each involving one of the two male nuclei from a single pollen tube.

- One male nucleus (n) fuses with the **egg cell nucleus** (n) to form a **diploid zygote** ($2n$), which develops into the embryo (plumule, radicle and cotyledons).
- The other male nucleus (n) fuses with the **two polar nuclei** ($n + n$) to form a **triploid endosperm nucleus** ($3n$), which divides repeatedly to form the **endosperm**, a food store (starch, protein, lipids) for the developing embryo.

After fertilisation the ovule develops into the seed and its integuments become the seed coat (testa); the ovary develops into the fruit.



Exam Tip: "Double fertilisation" does *not* mean two pollen grains or two eggs. One pollen tube delivers two male nuclei: one fertilises the egg (zygote, 2n), the other fuses with the two polar nuclei (endosperm, 3n). If a question asks you to *state* the ploidy of each, give the number (n, 2n or 3n) with no need for explanation; if it asks you to *explain* why the endosperm is triploid, say which three haploid nuclei fused.

Command Words and Common Misconceptions

Command word	What it asks for	Example in this topic
State / Give / Name	A short answer; no explanation needed	Name the enzyme that makes DNA from RNA in HIV
Describe	Say what happens or what something is like, in a logical order; no reasons needed	Describe what happens during anaphase I
Explain	Give reasons or mechanisms (use "because", "so", "therefore")	Explain why penicillin is more effective against Gram positive bacteria
Calculate	Get a numerical answer, showing working and units	Calculate the actual length of a cell from its image and magnification

Compare	Give similarities <i>and</i> differences, each point linked to both items	Compare spermatogenesis and oogenesis
Evaluate / Discuss	Consider strengths and weaknesses or both sides, then reach a judgement	Evaluate the use of untested drugs during the Ebola epidemic
Deduce / Suggest	Use the information given, or apply knowledge to a new situation	Suggest why a stained root tip shows few cells in anaphase

- **"Viruses are very small bacteria."** Viruses are not cells, have no metabolism and cannot be killed by antibiotics.
- **"Magnification and resolution are the same thing."** Magnification enlarges; resolution determines how much detail is visible.
- **"DNA is copied in prophase."** It is copied in S phase of interphase.
- **"Meiosis produces four identical cells."** The four haploid cells are genetically different.
- **"The egg is released as a mature ovum."** A secondary oocyte is released; it completes meiosis II only when a sperm enters.
- **"The pollen grain fertilises the egg."** A male nucleus, formed by mitosis of the generative nucleus, fertilises the egg.

Key Terms Checklist

Check that you can define and use each term:

Section	Key terms
2.1 Cells	☐ cell theory ☐ tissue, organ, organ system ☐ nucleoid ☐ plasmid ☐ 70S / 80S ribosome ☐ peptidoglycan ☐ Gram positive / Gram negative ☐ outer membrane (lipopolysaccharide) ☐ nucleolus ☐ rough and smooth ER ☐ Golgi apparatus ☐ lysosome ☐ centriole ☐ tonoplast ☐ magnification ☐ resolution ☐ TEM / SEM ☐ stain ☐ artefact ☐ eyepiece graticule ☐ stage micrometer

2.2 Viruses	☐ capsid ☐ envelope ☐ bacteriophage ☐ λ phage ☐ tobacco mosaic virus ☐ Ebola ☐ retrovirus ☐ reverse transcriptase ☐ provirus / prophage ☐ lytic cycle ☐ latency (lysogeny) ☐ antiviral ☐ contact tracing ☐ isolation / quarantine ☐ informed consent
2.3 Cell division	☐ interphase (G1, S, G2) ☐ sister chromatid ☐ centromere ☐ spindle ☐ prophase, metaphase, anaphase, telophase ☐ cytokinesis ☐ cell plate ☐ mitotic index ☐ meristem ☐ haploid / diploid ☐ homologous chromosomes ☐ bivalent ☐ chiasma ☐ crossing over ☐ independent assortment ☐ recombination ☐ translocation ☐ non- disjunction ☐ polysomy / trisomy ☐ monosomy ☐ Down's syndrome ☐ Turner's syndrome
2.4 Mammals	☐ spermatogonium ☐ primary / secondary spermatocyte ☐ spermatid ☐ oogonium ☐ primary / secondary oocyte ☐ polar body ☐ follicle ☐ capacitation ☐ zona pellucida ☐ acrosome reaction ☐ cortical reaction ☐ polyspermy ☐ zygote ☐ cleavage ☐ morula ☐ blastocyst ☐ trophoblast ☐ inner cell mass ☐ implantation
2.5 Plants	☐ anther ☐ pollen mother cell ☐ tetrad ☐ exine / intine ☐ tube nucleus ☐ generative nucleus ☐ male nuclei ☐ ovule ☐ micropyle ☐ megaspore mother cell ☐ embryo sac ☐ egg cell ☐ synergids ☐ antipodal cells ☐ polar nuclei ☐ pollen tube ☐ double fertilisation ☐ endosperm (3n)